Unlocking the Healing Potential

Cannabinoids in Spine Surgery for Pain Relief and Recovery

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Abstract

» Cannabinoids, such as D9-tetrahydrocannabinol and cannabidiol, interact with endocannabinoid receptors in the central nervous system and immune system, potentially offering pain relief. The entourage effect, resulting from the interaction of multiple cannabis components, may enhance therapeutic impact and efficacy, making them promising candidates for exploring pain relief in spine operations, known to be among the most painful operative procedures.

» The use of cannabinoids in pain management requires careful consideration of safety, including their cognitive and psychomotor effects, potential cardiovascular risks, risk of dependence, mental health implications, and drug interactions.

» Few studies have analyzed cannabinoid use in relation to spine surgery, with variable results reported, indicating possible effects on reoperation rates, mortality, complications, postoperative opioid use, and length of hospital stay.

» Current knowledge gaps exist in the understanding of cannabinoid effects on spine surgery, including the exploration of different administration routes, timing, dosage, and specific outcomes. In addition, mechanistic explanations for the observed results are lacking.

» Ethical considerations related to informed consent, medical expertise, societal impact, and legal compliance must also be thoroughly addressed when considering the utilization of cannabinoids in spinal pathologies and back pain treatment.

Since the 17th century, the cannabis plant has been a part of the American history from peacetime to wartime. It has been used for war supplies and recreation, with so-called "medical" uses tracing back to the 1800s. Federal legislation passed throughout the 19th century criminalized both "medical" and recreational marijuana use, going as far as to impose restrictions on academic marijuana research[1]. Since the last century, there has been a resurging interest in the utility of marijuana in medicine, including its applications in pain management.

Spine operations are ranked higher in association with elevated pain scores, similarly to other major orthopedic procedures, mainly because of insufficient pain management,[2,3] and most spine operations are elective[4,5]. In the field of spine surgery, achieving postoperative patient satisfaction poses a notable challenge, particularly when contrasted with other orthopedic procedures such as total hip surgery, knee arthroplasty, and shoulder surgery[6,7]. Although the latter procedures have reported satisfaction rates surpassing 90%, patients undergoing spine surgery have reported satisfaction rates in the 80s. One important factor contributing to these

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comparatively lower satisfaction rates in spine surgery is attributed to elevated preoperative expectations[6,7]. Swarup et al. have highlighted a correlation between higher expectations of postoperative pain relief and reduced postoperative satisfaction among spine surgery patients[7].

Notably, managing pain relief in spine surgery poses a substantial hurdle, particularly given the simultaneous rise in opioid abuse concerns.

In this article, we will examine the literature on marijuana use and its pain management properties to explore potential applications in spine surgery.

Mode of Action of Cannabinoids in Pain Mitigation

Cannabis is a plant that contains numerous chemical components, including 104 identified cannabinoids, and has attracted scientific interest due to the pharmacological properties of 2 specific cannabinoids: D9-tetrahydrocannabinol (D9-THC) and cannabidiol (CBD). Although D9-THC is psychoactive[8], CBD has garnered attention for its potential therapeutic effects. Endocannabinoids (or endogenous cannabinoids) and their receptors play a crucial role in regulating various physiological processes, including pain perception, neurotransmission, immune response, and bone health[9]. Cannabinoid receptors 1 and 2 (CB-1 and CB-2)[10] are the primary targets of the investigation, being G protein-coupled receptors activated by phytocannabinoids, endocannabinoids, and synthetic cannabinoid analogs. CB-1 is mainly expressed in the central nervous system (CNS) and peripheral nerve axons, primarily contributing to analgesic effects. CB-2, found in both the CNS and the immune system, is involved in neural protection against cell death[11]. Another receptor, G protein-coupled receptor 55[12], located in osteoblasts, osteoclasts, the gastrointestinal tract, and adrenal glands, interacts with endogenous cannabinoids resulting in intracellular calcium and diacylglycerol bioavailability.

Research conducted on human and rodent sensory neurons has shed light on the involvement of cannabinoid receptors in peripheral pain signaling. The activation of CB2 receptors has demonstrated the ability to reduce calcium ion influx triggered by capsaicin, leading to the desensitization of transient receptor potential cation channel subfamily V member 1 (TRPV1) receptors responsible for pain sensation[13]. Studies in guinea pig and rat models have similarly indicated that cannabinoids, acting through CB2 receptors, can influence sensory nerve activity, offering a potential mechanism for pain regulation[14-16]. Moreover, cannabinoids, particularly CBD, interact with other receptors such as TRPV1, which plays a role in thermal hyperalgesia and integrates various pain-inducing stimuli[13]. An endogenous cannabinoid, anandamide, activates both cannabinoid and TRPV1 receptors, thereby inhibiting the release of neuropeptides associated with inflammatory pain[17]. These collective findings suggest that cannabinoids have the potential to modulate chronic pain through synergistic action on multiple targets[18].

Entourage Effects

Cannabis plant contains a complex mixture of chemical compounds called cannabinoids, terpenes, and flavonoids[19]. The interaction of these components gives rise to what is known as the "entourage effect"[20]. The entourage effect refers to the synergistic interaction among these various compounds, resulting in a more substantial therapeutic impact when consumed together compared with the isolated use of individual cannabinoids. Understanding this phenomenon is crucial in comprehending the therapeutic potential of cannabis and optimizing its medicinal applications[21].

The entourage effect is believed to result from multiple mechanisms. First, cannabinoids and terpenes may act on different receptors within the endocannabinoid system (ECS), leading to a more comprehensive modulation of physiological processes. For example, CBD can attenuate some of the adverse effects of THC, such as anxiety and tachycardia, by interacting with different receptor sites[22-25]. Second, terpenes can enhance the bioavailability of cannabinoids by promoting their absorption through various biological membranes. Moreover, some terpenes are believed to inhibit specific enzymes responsible for metabolizing cannabinoids, thereby prolonging their effects[26].

When cannabinoids and terpenes are present together in cannabis products, they can potentially offer a more well-rounded and effective treatment approach for various conditions, including chronic pain, anxiety, chemotherapy-induced nausea, and epilepsy[27]. A randomized trial investigating the efficacy of a mixture of compounds compared with a single compound in patients with chronic, central neuropathic pain from brachial plexus injury found improvements in pain score and sleep[28], and favorable effects in alleviating central neuropathic pain in patients with multiple sclerosis[29]. These holistic interactions potentially suggest that whole-plant cannabis preparations rather than isolated cannabinoids may provide a mixture of molecular compounds that can act synergistically to alleviate symptoms.

Despite the growing recognition of the entourage effect, its full extent and mechanisms remain subject to ongoing research. The wide variety of cannabis strains, each containing a unique combination of cannabinoids and terpenes, makes it challenging to standardize and predict the exact outcomes of specific cannabis products. In addition, constraints such as regulatory restrictions, limited access to study materials, and funding challenges have impeded comprehensive investigations into this complex phenomenon[30].

Safety Considerations

The utilization of cannabinoids in the healthcare sector, including their

application in pain management and enhancement of postoperative outcomes in spinal surgery, must be accompanied by a rigorous evaluation of safety considerations. There are several potential risks and side effects associated with their use, as well as potential drug interactions that may affect the safe implementation of cannabinoid-based therapies.

Cognitive and Psychomotor Performance

Cannabinoids, notably THC, are known to affect cognitive and psychomotor performance. These effects may manifest as short-term memory impairment, altered judgment, and slowed reaction time[31-33]. Although these impacts are usually transient and can diminish with continued use, they are of particular concern to individuals who need to maintain sharp cognitive function and psychomotor performance for their jobs or daily activities. In the context of the postoperative period after spine surgery, patients are often expected to participate in rehabilitation exercises, which can include physical therapy and occupational therapy. Cognitive and motor impairments could substantially affect the patient's ability to fully engage in these activities, potentially prolonging recovery time and affecting the overall success of the surgical intervention. Furthermore, it is essential to consider the increased risk of accidents and falls associated with these side effects, which could be severely detrimental during the postoperative recovery period.

Cardiovascular Effects

THC can induce dose-dependent cardiovascular effects, including tachycardia and hypertension. In isolated cases, it can even provoke more severe cardiovascular events such as myocardial infarction and stroke, especially in patients with underlying cardiovascular conditions[34]. For spine surgery patients who often have comorbid conditions, this risk needs to be closely monitored. Older adults and those with known cardiovascular disease are particularly at risk. A thorough cardiovascular risk assessment should be conducted with cannabinoids used in postoperative pain management in spine surgery.

Dependence and Withdrawal

Cannabinoids, despite their potential therapeutic benefits, have a risk of fostering physical dependence[18]. When cannabinoid use is abruptly discontinued, withdrawal symptoms can occur. These symptoms include restlessness, irritability, mild agitation, insomnia, sleep disturbances, nausea, and cramping[35-37]. Medical practitioners must be vigilant about the dosage and duration of cannabinoid administration, and gradual tapering might be necessary when discontinuing cannabinoid use to mitigate the withdrawal effects.

Mental Health

Mental health implications are also important to consider when using cannabinoids, especially THC, for pain management. High doses of THC can induce symptoms such as anxiety, paranoia, and even hallucinations. It may also exacerbate pre-existing mental health disorders, such as depression, bipolar disorder, and schizophrenia[38,39]. Mental health screening and continuous monitoring should be an integral part of postoperative care when cannabinoids are used.

Interactions with Other Illicit Substance Abuse

Patients with recreational marijuana or cannabinoid use disorder often exhibit a higher likelihood of having concurrent substance use disorders, potentially attributable to their similar risk factors or genetic predisposition[40]. Among these concurrent disorders, alcohol and nicotine use, particularly in the form of tobacco consumption, are the most prevalent[40]. Although the individual effects of alcohol and nicotine use are well-documented, data regarding their couse are still limited. A study investigating the simultaneous use of tobacco and marijuana revealed that, although the risk of complications was elevated in both the tobacco-only and tobacco-marijuana combination groups compared with cannabis use alone, the risk was notably higher in the tobacco-only group when compared with the combination group[41]. This intriguing finding supports the hypothesis that marijuana may enhance nicotine levels in the body, potentially aiding in smoking cessation[42]. Moreover, this observation may elucidate the influence of marijuana use on tobacco consumption and its surgical implications. Notably, the route of drug administration plays a crucial role. Concurrent tobacco and marijuana usage has been associated with compromised lung function and increased risk of respiratory complications[43]. Therefore, when patients present for spine surgeries, their lung function should be considered to mitigate the risk of postoperative respiratory complications.

CBD, acting as a serotonin 1A receptor agonist, exhibits potential in reducing alcohol consumption and shielding against alcohol-related harm such as liver and brain damage[44]. Clinical studies reveal that combining CBD with alcohol impairs motor functions but enhances self-awareness regarding intoxication and deficits[45]. CBD may also lower blood alcohol levels. Although the interactions of the ECS with alcohol are well-established, their combined impact on surgical outcomes remains insufficiently studied, warranting further exploration considering their common coexistence.

Drug Interactions

Given that cannabinoids are primarily metabolized by cytochrome P450 enzymes, this makes them susceptible to interactions with other medications metabolized through the same pathway[46] (Table I). Depending on the medication, it may either inhibit or induce the activity of these enzymes, which can subsequently affect the levels and efficacy of cannabinoids within the body. The concurrent use of

TABLE I Interactions Between Cannabinoids and Common Medications in Spine Surgery*

Drug Class	Mode of Interaction	Effect
Sedatives	Concurrent use with CNS depressants (e.g., opioids and benzodiazepines)	Enhanced sedative effects Excessive sedation Impairment of cognitive and motor functions Increased risk of falls Potential respiratory depression
Antidepressants and antipsychotics	THC may elevate levels of certain antidepressants and antipsychotic medications	Enhanced effects of antidepressants Associated side effects may be intensified
	CBD may decrease efficacy of certain antidepressants and antipsychotics	Reduced effectiveness of antidepressants Potential antagonistic effects
Anticoagulants	Interference with anticoagulant activity (e.g., warfarin)	Increased risk of bleeding Elevated anticoagulation levels Potential higher risk of bleeding

*CBD = cannabidiol, CNS = central nervous system, and THC = tetrahydrocannabinol.

cannabinoids and other central nervous systems' depressants, such as opioids and benzodiazepines, can enhance sedative effects. This can lead to excessive sedation, impairment of cognitive and motor functions, and increased risk of falls. In more severe cases, coadministration of these medications may lead to respiratory depression, a potentially life-threatening condition[47,48]. Considering these interactions or potentially selecting an alternative treatment option may be necessary. Interaction with certain antidepressants and antipsychotic medications is also a possibility. In some cases, high doses of THC may elevate the levels of these drugs within the body, potentially enhancing their effects and associated side effects. CBD, on the other hand, could decrease the efficacy of these medications by antagonizing the receptors they bind to[34,49]. Cannabinoids may interfere with the activity of anticoagulants, such as warfarin, leading to increased anticoagulation and a potentially higher risk of bleeding[50]. This risk may be exacerbated in a postoperative context where the risk of bleeding may already be elevated.

When coadministering cannabinoids and anticoagulants, frequent monitoring of blood coagulation parameters is warranted. A dose adjustment of the anticoagulant may also be required, and the patient should be educated about the signs of excessive anticoagulation, such as prolonged bleeding from minor cuts, blood in urine or stool, and unexplained bruising.

Clinical Evidence and Studies

Thus far, there have not been many studies analyzing marijuana use in relation to spine surgery. What work there has been thus far has been minimal and has not given any definitive answers[51], thus calling for larger future studies with the hopes of future clinical applications (Table II). Overall, the variable results have shown cannabinoids to possibly increase the risk of reoperation[52,53], both decrease mortality and increase risk of complications[54,55], both increase and decrease postoperative opioid use[56,57], increase and decrease length of hospital stay for spine surgery[55,58,59], and both increase[60] and reduce pain in patients with general spinal pathology, likely due to overall reductions in postoperative pain[61-63]. However, the evidence remains contradictory regarding cannabinoids and their ability to treat postoperative pain after spinal cord injuries as shown by a recently published meta-analysis[60]. Heard et al.[51] also demonstrated, that for every study supporting CBD/THC, there seems to be another against it for the exact same parameters.

Owing to marijuana's illicit nature in many places around the world, including many states in the United States of America, where it remains illegal on the federal level, most of the studies regarding its use have been retrospective surveys and cohorts that seek to see how these patients fared in comparison with nonusers[52,56-58,64]. Thus, there have been very few studies looking at CBD/THC use after surgery and even fewer that assess postoperative pain in marijuana users with access to marijuana in the postoperative period in the hospital. The few trials, there have been[61,62], have involved very small sample sizes and are not recent. Spine surgery has rarely been isolated in the study of cannabinoids and is often grouped with other types of surgery such as gynecologic surgery[65].

The current gaps in the knowledge of how cannabinoids may affect spine surgery are immense. The literature does not account for different routes of administration, the timing of marijuana use in relation to spine surgery, or the amount used. It also has not shown any definitively different outcomes between CBD, THC, or mixtures of the 2 compounds. With drug use of any kind, especially illicit use, confounding variables must be accounted for, and many studies fail to do this. Importantly, studies lack mechanistic explanations for the results that have been found regarding reoperation rates[53],

TABLE II Summary of Studies Reviewing the Impact of Marijuana or Cannabinoid Use in Spine Surgery*

Year	Author	Study Design	Sample Size	Surgery	Form/Route	Outcome
2023	Heard et al.[51]	Review	—	—	—	—
2023	Silver et al.[56]	Retrospective case-control	2,678	ACDF	Self-reported	Less postop opioid use
2022	D'antonio et al.[53]	Retrospective cohort	259	1 or 2-level lumbar fusion	Self-reported	Increased risk of reoperation
2022	Lambrecths et al.[52]	Retrospective cohort	240	ACDF	—	Increased risk of reoperation
2022	Moon et al.[57]	Retrospective review	—	Lumbar fusion	Preoperative diagnosis as cannabis user	More postoperative opioid use
2022	Jakoi et al.[59]	Retrospective cohort	294	Lumbar fusion with TXA	Self-reported inhalation and ingestion	Shorter length of stay
2021	Chiu et al.[55]	Clinical case series	423,978	Elective spine surgery	Self-reported	Increased perioperative morbidity
2020	Jakoi et al.[58]	Retrospective cohort	36 (106)	TLIF	Self-reported	Shorter length of stay
2019	Moon et al.[54]	Retrospective cohort	Not only spine†	Self-reported	Mixed outcomes	—
2019	Liu et al.[64]	Retrospective cohort	3,793	Spine surgery	—	More pain on movement
2018	Mondello et al.[63]	Retrospective	11	n/a (failed back)	Oral	Pain in failed back syndrome
2016	Yassin et al.[62]	Cross-over study	9	Fusion	Smoking	Lumbar ROM, LBP, sciatica reduction
2006	Beaulieu[65]	Pilot trial	—	—	—	More pain

*ACDF = anterior cervical discectomy and fusion, LBP = low back pain, ROM = range of motion, TLIF = transforaminal lumbar interbody fusion, and TXA = tranexamic acid.
†This study evaluated the impact of CBD use in various orthopedic procedures and did not exclusively focus on spinal procedures.

postoperative mortality[54], and the perceived experience of pain[55].

Ethical Considerations and Patient Autonomy

As the utilization of marijuana or cannabis for spinal pathologies and back pain gains traction, several ethical considerations arise.

Informed Consent

Owing to the complex legal and social status of cannabis, ensuring informed consent becomes challenging. Healthcare providers must inform patients about the potential benefits and risks of cannabis use while adhering to regional regulations and professional guidelines[66]. Patients should be educated about the limited clinical evidence available, as well as the lack of standardized dosing and formulation of cannabis products, allowing them to make fully informed choices.

Medical Professionalism and Expertise

The ethical use of cannabis for spinal pathologies and back pain requires a well-informed healthcare provider. Physicians and healthcare professionals must remain up to date with the latest research and evidence regarding cannabis-based treatments. Medical professionals must understand the potential interactions between cannabis and other medications, especially considering the high prevalence of polypharmacy among patients with chronic pain[67].

In addition, providers should be knowledgeable about potential adverse effects and be prepared to counsel patients appropriately. Even in countries where medical, as well as recreational use of cannabis, is legalized, physicians feel uncomfortable with their expertise in prescribing it for medicinal use[68,69]. They could benefit from education and training in the subject. Healthcare providers, especially spine surgeons who often treat patients with increased pain management needs, should be knowledgeable about the 3 Food and Drug Administration–approved synthetic cannabinoid medications.

Societal Impact and Stigma

The cultural and societal perception of cannabis use continues to evolve, but it remains a controversial topic[70,71]. Patients using cannabis for spinal pathologies may encounter stigma from family members, colleagues, or even healthcare providers who hold strong opinions about marijuana use. Medical

professionals must create a nonjudgmental and compassionate environment, ensuring patients feel comfortable discussing their treatment choices openly[72]. Moreover, it is essential to advocate for unbiased research and public education to dispel myths and misunderstandings surrounding cannabis use.

Legal Implications and Compliance

The legality of cannabis use varies significantly between countries and regions, ranging from complete prohibition to medical or recreational legalization[73]. This legal landscape poses ethical challenges for healthcare providers who must adhere to local regulations while addressing patients' needs for pain relief. On July 7, 2014, New York enacted the Compassionate Care Act, joining 22 other states in legalizing medical marijuana (MMJ). As of November 2017, MMJ laws are in place in 29 states along with Washington, DC. In addition, 17 more states have established laws specifically for CBD, and 8 states, along with DC, have legalized recreational cannabis use[74]. By contrast, at the federal level, botanical cannabis and its components remain prohibited for medical purposes and are classified as Schedule I substances under the Controlled Substances Act. This classification designates them as substances with the highest potential for harm and without recognized medical benefits[75].

It is essential for the healthcare providers to understand the limitations surrounding prescribing or legally dispensing botanical cannabis or its components, which can only occur within federally approved research programs[76,77]. In states with MMJ laws, such as New York, physicians are allowed to "certify" patients for qualifying conditions and "recommend" MMJ use, but they lack the authority to issue prescriptions. Physicians who wish to certify patients must undergo a 4-hour course covering the ECS, physiological and adverse effects of cannabinoids, and dosing and administration guidelines[78,79].

Conclusion

The limited evidence regarding marijuana use in spine surgery is often hindered by its illicit status which results in retrospective studies rather than comprehensive investigations. Ethical considerations further complicate the integration of cannabinoids into clinical practice. Despite the growing interest in cannabinoids for pain management in spine surgery, the literature remains inconclusive and lacks mechanistic explanations for the varied outcomes reported. The complexity of marijuana's components, different routes of administration, timing, and dosage makes standardization challenging. Addressing these gaps in knowledge requires conducting larger, well-controlled studies that consider the various aspects of cannabinoid use in spine surgery. The legal landscape, however, poses a major challenge to conducting rigorous, randomized control trials involving marijuana. The authors support Senators Cornyn and Feinstein, who in 2021 released a bipartisan statement encouraging the lifting of federal barriers on marijuana research to better understand its health impact[80]. Since their statement, the bipartisan Medical Marijuana and Cannabidiol Research Expansion Act has become law[81], and we hope this will unleash a wave of high-quality research on marijuana and pain control in orthopedic spine surgery. As the evidence unfolds, healthcare professionals should educate patients, stay up to date on the latest research, and understand potential interactions and adverse effects of marijuana in controlling pain. Societal perception and legal considerations must be taken into account while advocating for unbiased research and public education.

In summary, although cannabinoids hold promise in pain management and recovery after spine surgery, further research is essential to establish their safe and effective implementation in clinical practice. The path ahead requires collaboration between researchers, healthcare professionals, and policymakers to unlock the full potential of cannabinoids in improving patient outcomes and ensuring their responsible use in the medical context.

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